Assignment #5.1: Midterm. Supervised Learning with k-NN Classification

Think of your own possible problem that could be solved with the specific Supervised Learning method: **k-NN Classification algorithm**.

1. Describe the problem, the data which you could use to train and eval the model.

I would like to use **k-NN Classification** to predict whether a customer will subscribe to a bank term deposit after a marketing campaign. This is a **binary classification problem** because the result is either **yes** or **no**.

I would use the **Bank Marketing Dataset (bank-additional-full.csv)** from the UCI Machine Learning Repository (https://archive.ics.uci.edu/dataset/222/bank%2Bmarketing?utm_source=chatgpt.com). The dataset includes customer and campaign information, such as age, job, education, account balance, loan status, contact type, and previous campaign outcome. The target variable is y, which shows whether the customer subscribed.

I would split the data into training (70%), validation (20%), and test (10%) sets. The model can help banks identify customers who are more likely to subscribe to the bank term deposit and improve marketing efficiency.

2. Describe the input features, their type and data pre-processing.

The input features include customer information, campaign information, previous campaign information, and economic indicators.

The numerical features include age, duration, campaign, pdays, previous, emp.var.rate, cons.price.idx, cons.conf.idx, euribor3m, and nr.employed.

The categorical features include job, marital, education, default, housing, loan, contact, month, day_of_week, and poutcome.

Before using k-NN, I would preprocess the data because k-NN is distance-based. I would use one-hot encoding to convert categorical variables into dummy variables, scale numerical features so they are on a similar range, and convert the target variable y into binary labels, where yes = 1 and no = 0. For unknown values in categorical features, I would first keep them as a separate category instead of treating them as missing values, because unknown may still contain useful information about the customer or campaign record.

3. Describe your approach on choosing the details (hyperparams) of k-NN Classification model; do we need the training process and what are the loss functions to use during it?

I would start with k=7 as an initial choice because it is a moderate value and can reduce the chance of overfitting. Then I would test other values, such as k = 3, 5, 9, 11, and 15, on the validation set and choose the value with the best performance. For this problem, I think Manhattan distance may work well because the dataset contains scaled numerical features and one-hot encoded categorical features, but I would compare it with Euclidean distance later. For voting, I would first try distance-based voting because closer customer records may be more relevant, but I would also compare it with uniform voting.

k-NN needs training data, but it does not have a traditional training process. It mainly stores the processed training data, and most computation happens during prediction. Therefore, k-NN does not use a loss function during training. I would evaluate the model using a confusion matrix, accuracy, precision, recall, and F1-score.

4. Describe your own approach to selecting the best model.

I would use **5-fold cross-validation** to compare different combinations of **k**, **distance metric**, and **voting** method. For each combination, I would calculate accuracy, precision, recall, F1-score, and confusion matrix. Since the dataset may be imbalanced, I would focus more on **F1-score** and **recall for the yes class** instead of relying only on accuracy. After selecting the best model setting, I would evaluate it once on the separate test set.

5. Describe the approach to the iterative model and data update.

For the iterative model and data update, I would use a **sliding-window update strategy**. For example, when 1,000 new labeled customer records are collected from a new campaign, I may add them to the stored training dataset and remove the oldest or least relevant 1,000 records. This can help keep the k-NN reference data closer to current customer behavior and also control prediction cost.

I would also consider using **sample weights**. Recent records or records from similar campaigns could receive higher weights because they may be more relevant to the current prediction task. However, I would validate this strategy before applying it, because giving too much weight to recent data may cause the model to overreact to short-term changes. If the updated dataset and weighting strategy improve F1-score and recall for the yes class, I would use the updated version.

6. Describe how you would **track** the possible future issues in the model prediction quality and how to **improve** them.

To track future prediction quality, I would regularly monitor the confusion matrix, accuracy, precision, recall, and F1-score. Since the dataset may be imbalanced, I would focus more on recall and **F1-score for the yes class** instead of relying only on accuracy.

I would also check data drift by comparing new campaign data with the original training data, such as changes in customer age, job type, loan status, contact method, economic indicators, and the **yes/no** ratio.

To improve the model, besides adding recent labeled data and removing outdated or low-quality records, I also would like to re-tune hyperparameters such as **k, distance metric, and voting method**. I would also review false positives and false negatives to understand where the model makes mistakes.